

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASWY")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,           TruncNormal,              0.1,           0.2,           0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,          0.8,           0.1,           0.1, 2);
Distrib (kap,          Uniform,                  0.1,           0.9);
Distrib (Ehp,          TruncNormal_cv,          100,           0.2,           40, 300);
Distrib (E_coc,        TruncNormal_cv,          30,            0.2,           1, 300);
Distrib (rOM_ClxHorse, LogUniform,              0.00001,       0.3);
Distrib (w, TruncNormal_cv,          40, 10,          5, 100);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcaliSWY DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	133.712000	0.36823800	0.87007400	0.64470600	131.55800	11.132700
2.5%	123.935000	0.31415700	0.78721300	0.59052325	102.03200	8.239337
97.5%	158.276700	0.56854400	1.10411000	0.66100100	177.32600	14.632000
Min	83.184900	0.04100250	0.61814500	0.17816500	63.11510	6.614340
Max	173.771000	0.76357700	1.17569000	0.68772800	215.91400	29.348300
Result_mean	141.719678	0.41467859	0.93433066	0.62823358	142.04077	11.210939

Result_sd	8.576233	0.06373704	0.08014498	0.01780998	18.16518	1.598371
	rOM_ClxFhorse.1.	w.1.	wv.1.	Sigma_W.1.		
Result_mode	0.0000134155	19.60560	98.23120	0.15421600		
2.5%	0.0000256131	14.95230	25.34322	0.13085200		
97.5%	0.0081660500	27.62015	98.71680	0.16814100		
Min	0.0000102204	11.51290	1.50073	0.11510700		
Max	0.0509530000	86.31360	99.99190	6.06027000		
Result_mean	0.0034963771	20.64595	69.19801	0.14926241		
Result_sd	0.0021228160	3.29933	20.81524	0.04794719		

Number of observations, n = 277
Number of parameters, k = 9
Log-likelihood, LL = -50.55463
BIC = 151.7254

Potential scale reduction factors:

	Point est.	Upper C.I.
E_coc.1.	1.30	1.90
Ehp.1.	1.22	1.64
Fm.1.	1.03	1.05
kap.1.	1.03	1.05
pAm.1.	1.17	1.48
r_pAm_pM.1.	1.17	1.53
rOM_ClxFhorse.1.	1.01	1.04
Sigma_W.1.	1.00	1.01
w.1.	1.31	1.94
wv.1.	1.00	1.02

Multivariate psrf

1.21

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

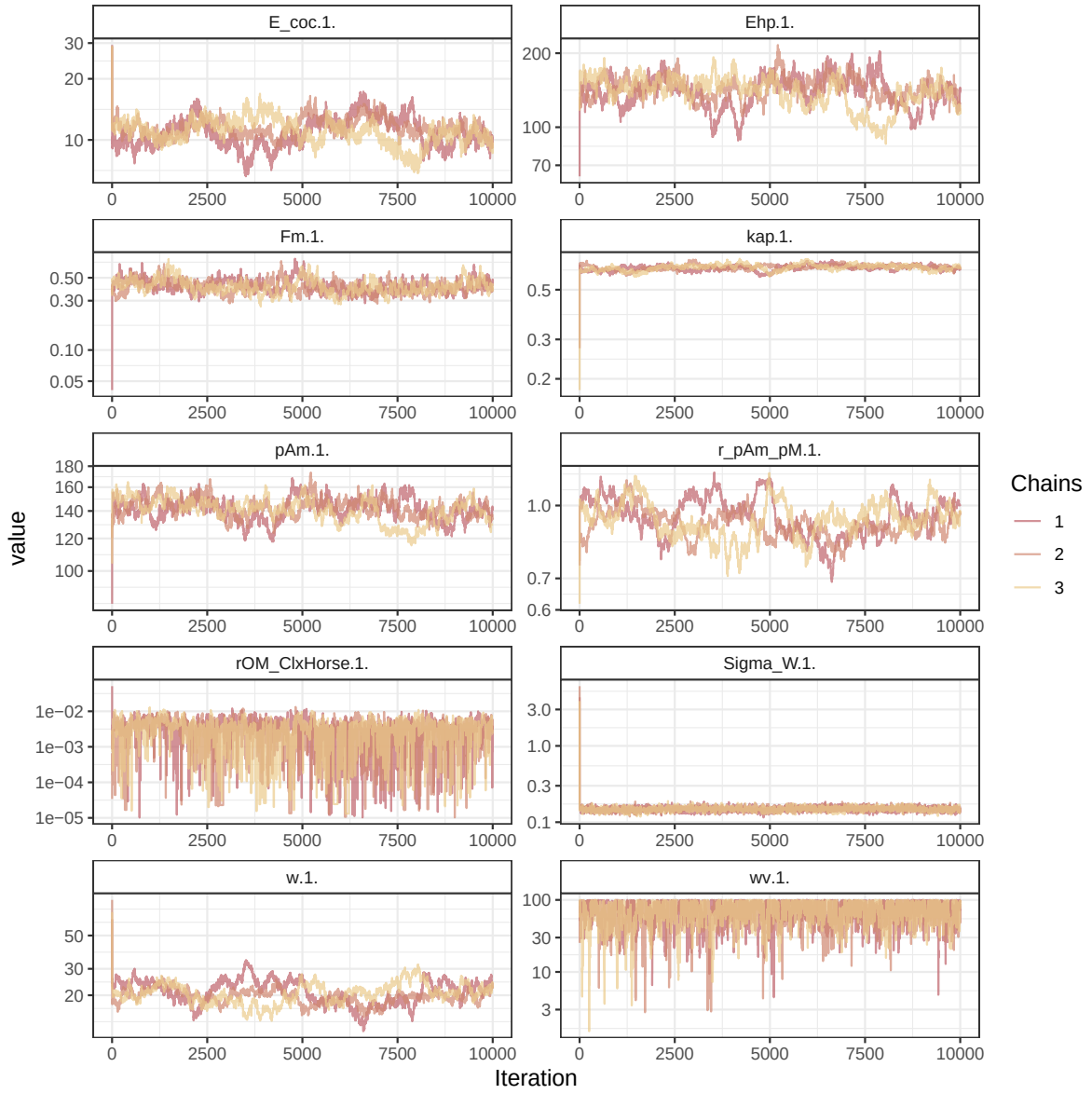


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

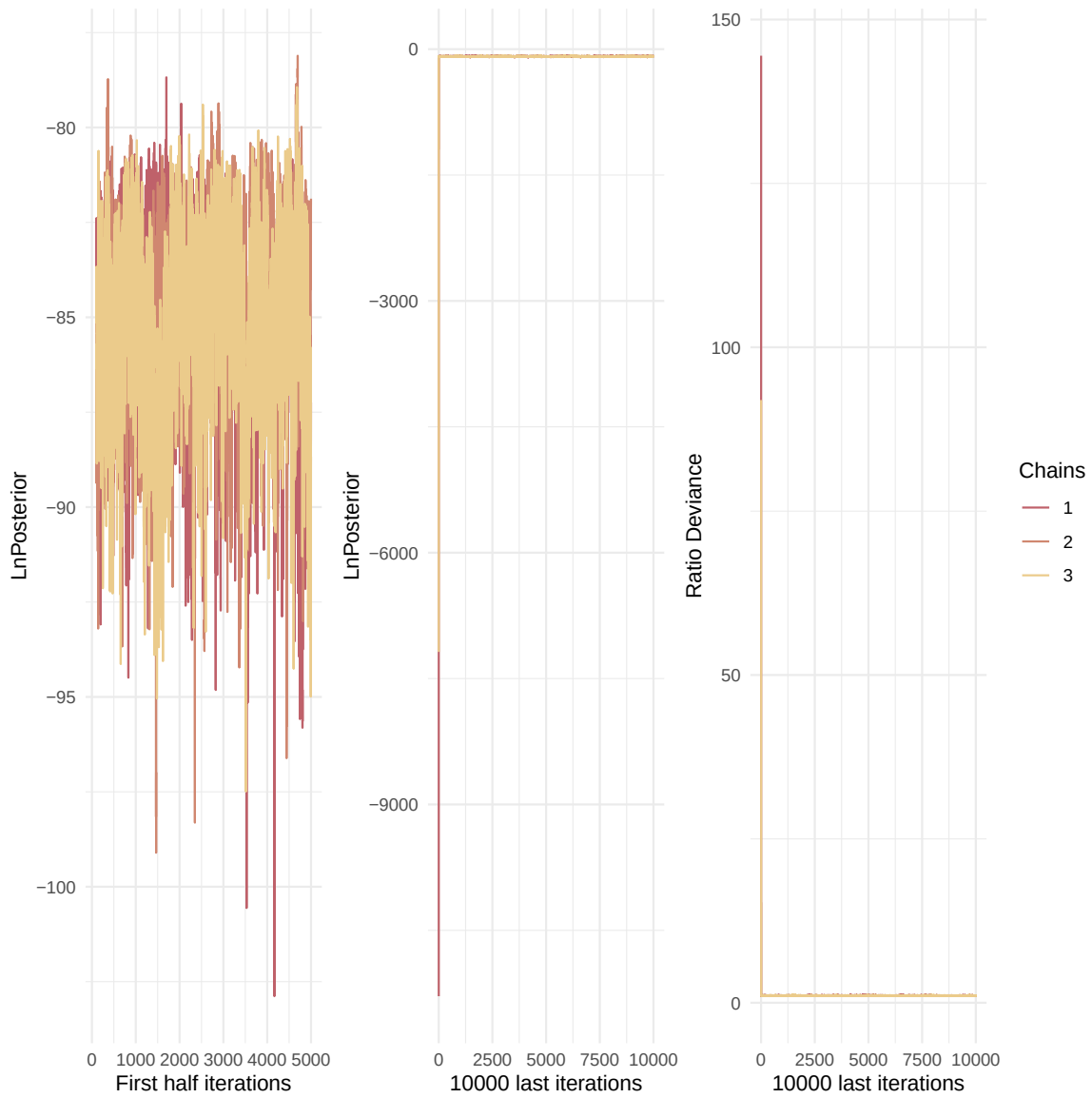


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

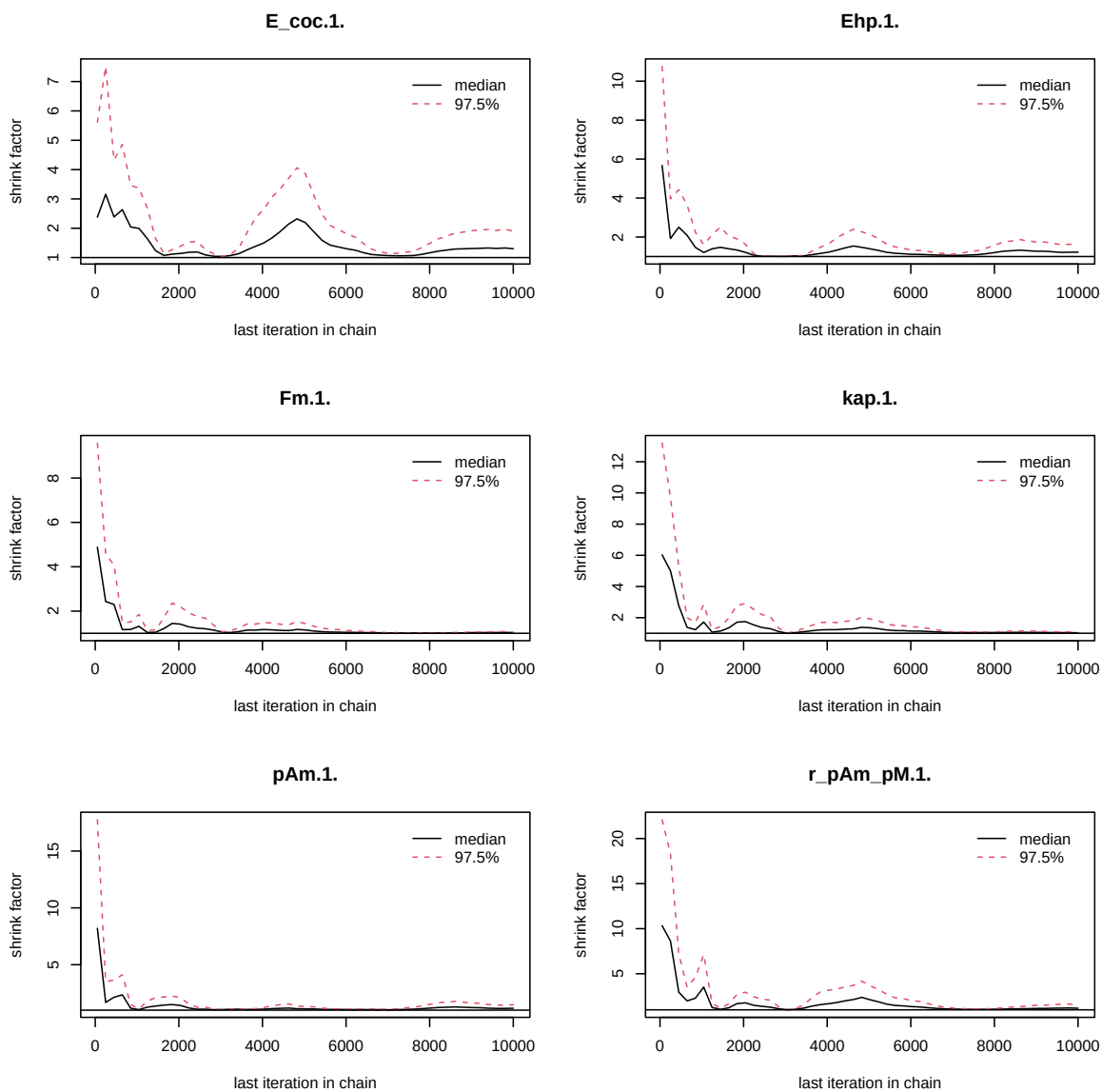


Figure 3: Chains convergence.

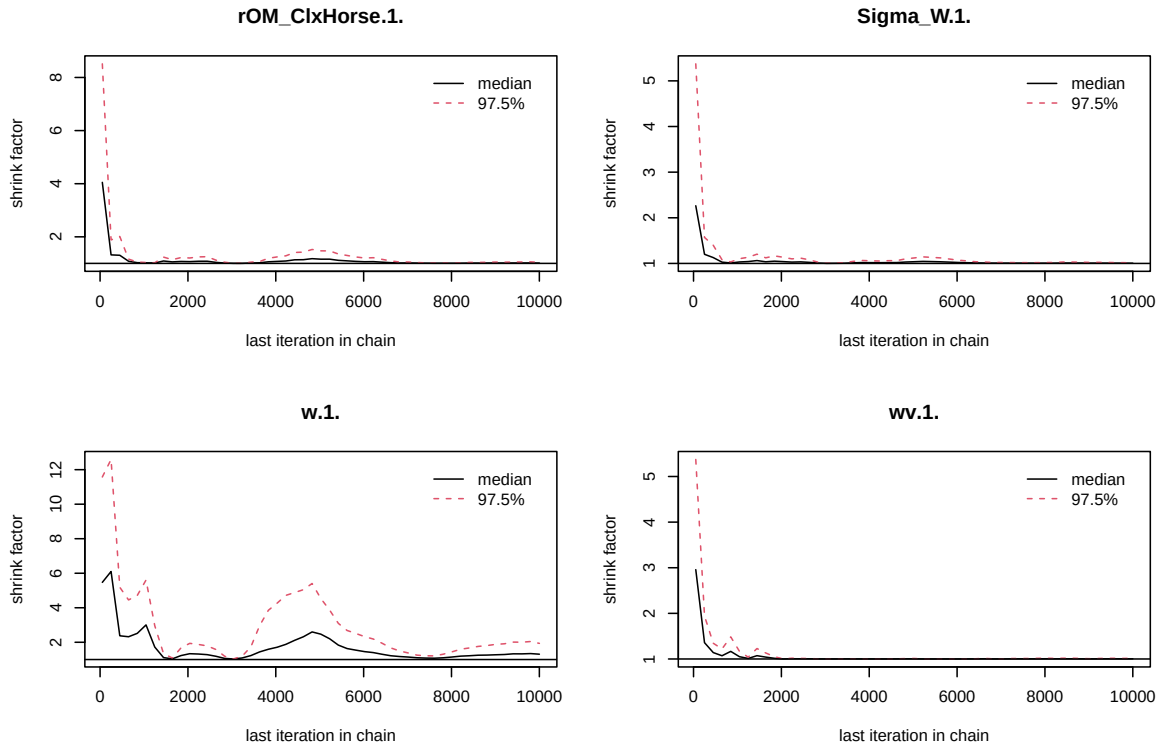


Figure 4: Chains convergence.

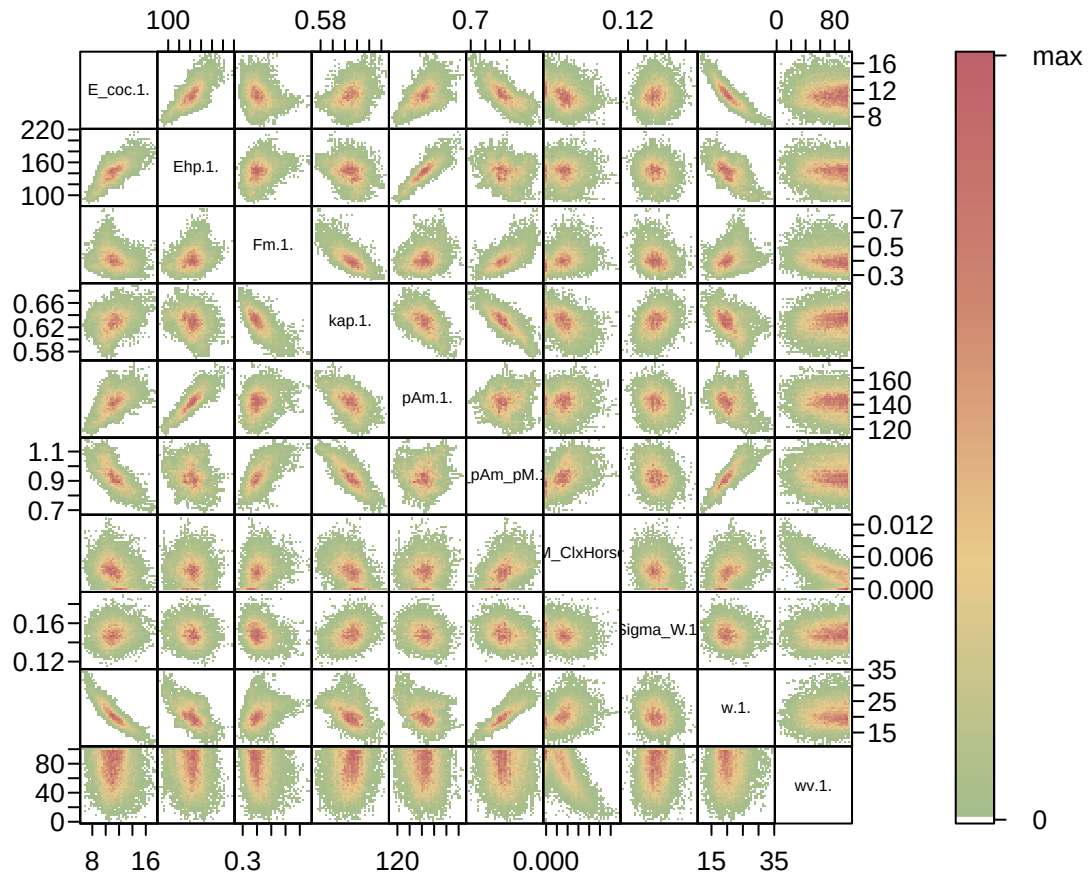


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```


**Posterior
and prior distributions**

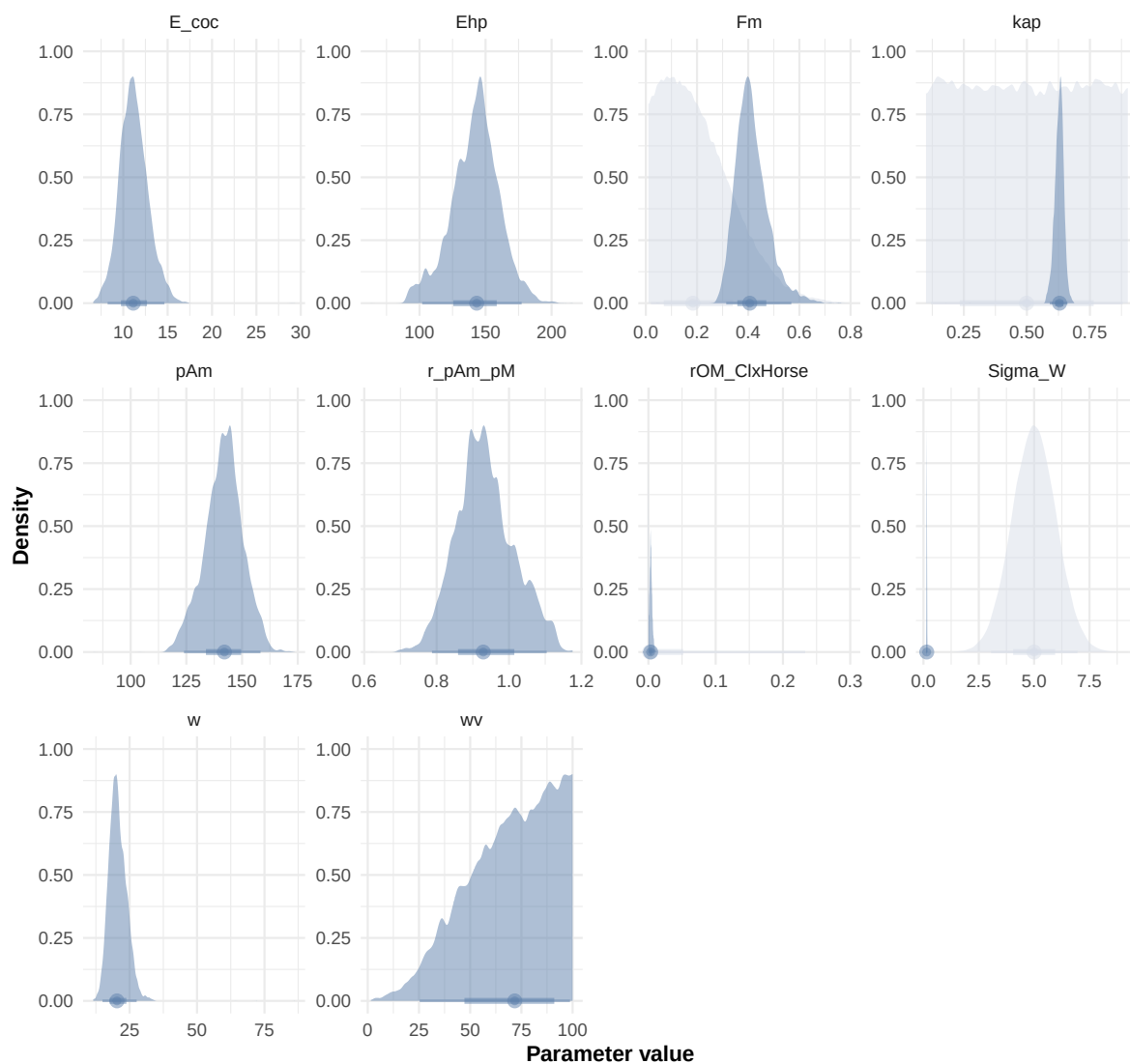


Figure 6: Distributions of the priors and the posteriors.

```
"Bart2020",  
#"Gollot2026_lufa"  
"Gollot2026_dens_D1",  
"Gollot2026_dens_D2",  
"Gollot2026_dens_D2b",  
"Gollot2026_dens_D5",  
"Gollot2026_dens_D7"  
)  
  
Adults_alone <- FALSE  
  
df_data <- f_import_data_DEB(l_Experiments, Adults_alone)  
  
text_param <- Summary_res["Result_mode", , drop = FALSE] %>%  
  t() %>%  
  as.data.frame() %>%  
  tibble::rownames_to_column("param") %>%  
  rename(val = 2) %>%  
  mutate(  
    param = gsub("\\\\.1\\.$|\\.1\\.\"", "", param),  
    line = glue("{param} = {val};")  
  ) %>%  
  pull(line) %>%  
  paste(collapse = "\n")  
  
Endpoints_print <- "Weight,  
  Reproduction"  
  
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)  
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr  
  
# ./mcsim.DEBcaliSWY DEB_sim_full.in  
  
#####  
df_No_sim <- df_data |>  
  dplyr::select(ID_experiment, No_sim)  
  
df_carac_color <- data.frame(  
  carac = unique(l_carac),  
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],  
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])  
)
```

```

v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between 'x' and 'y'.
 i Row 1 of 'x' matches multiple rows in 'y'.
 i Row 1 of 'y' matches multiple rows in 'x'.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

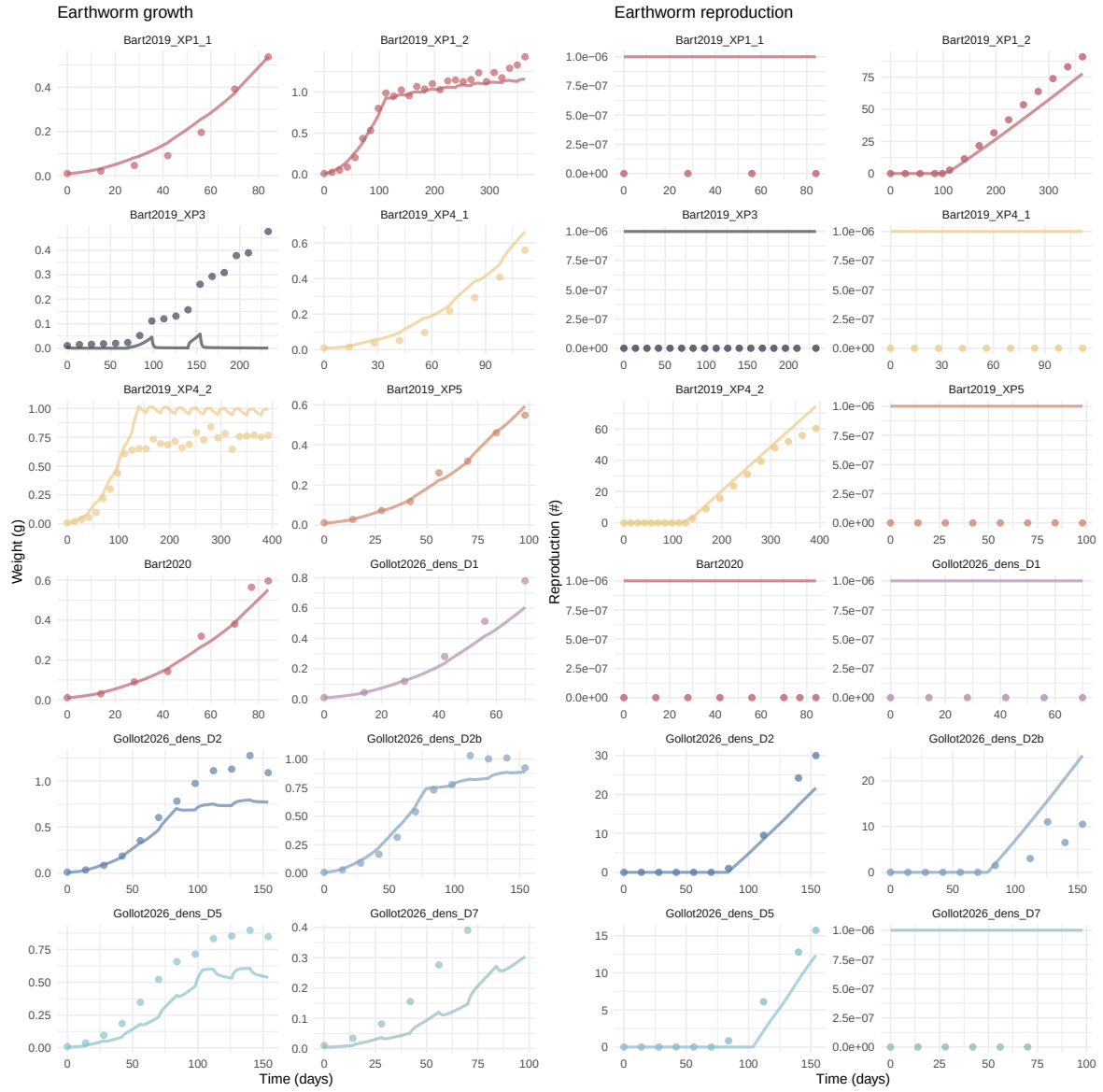
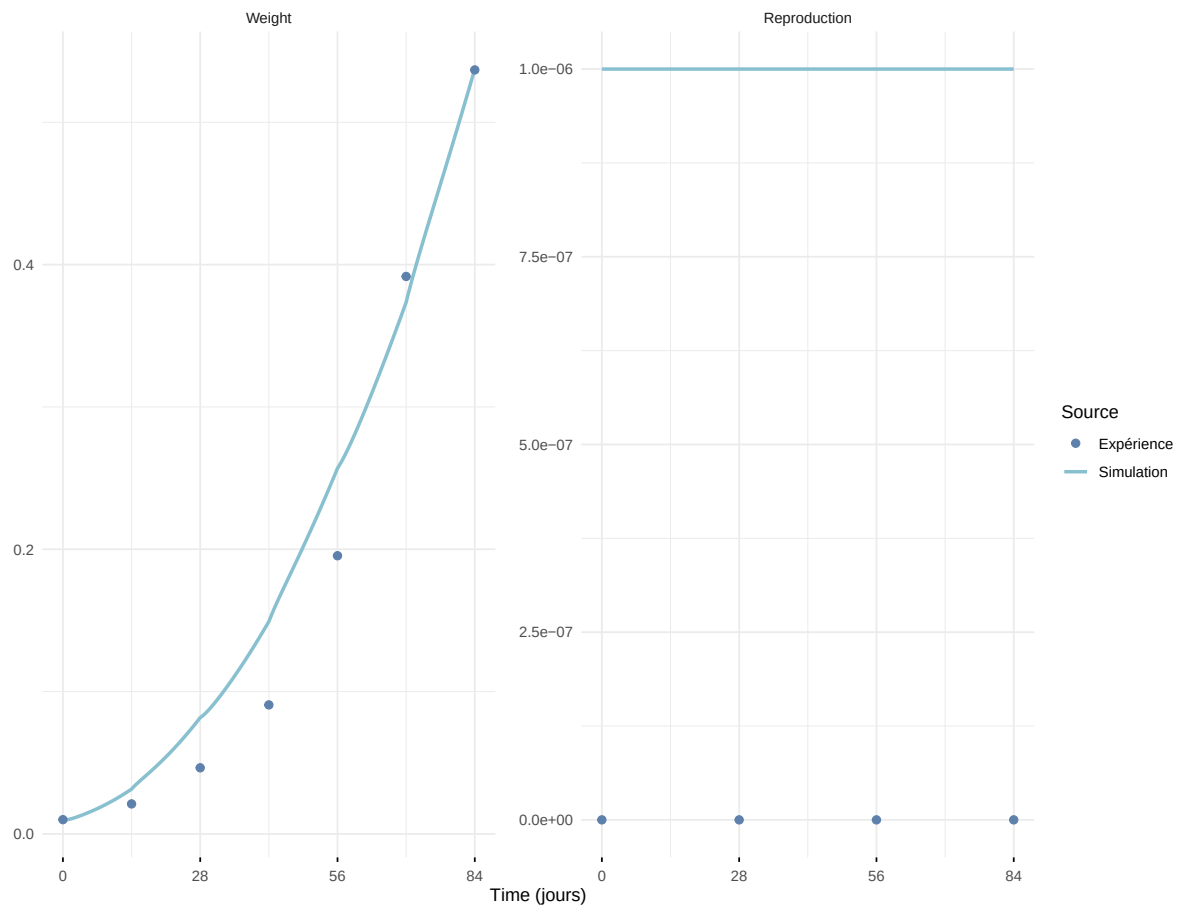
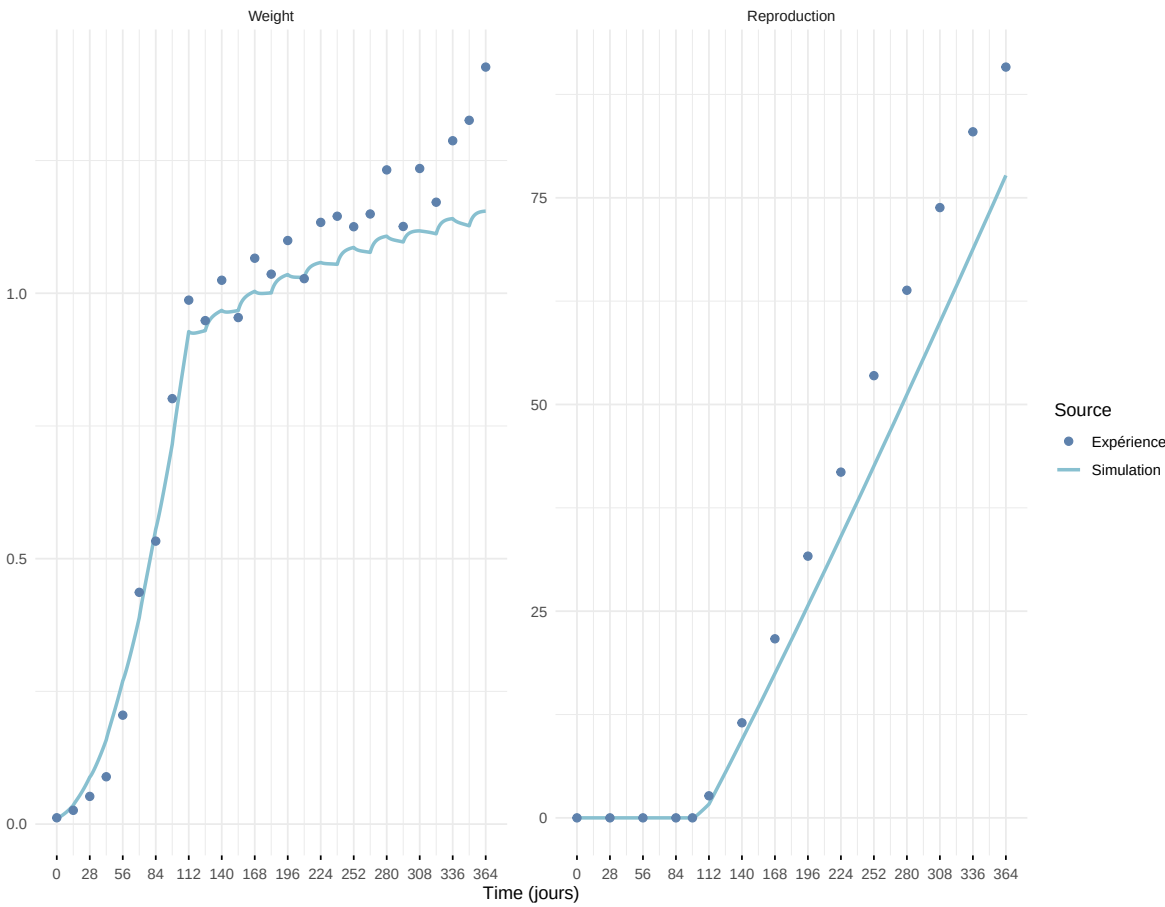


Figure 7: Simulations Weight and Reproduction.

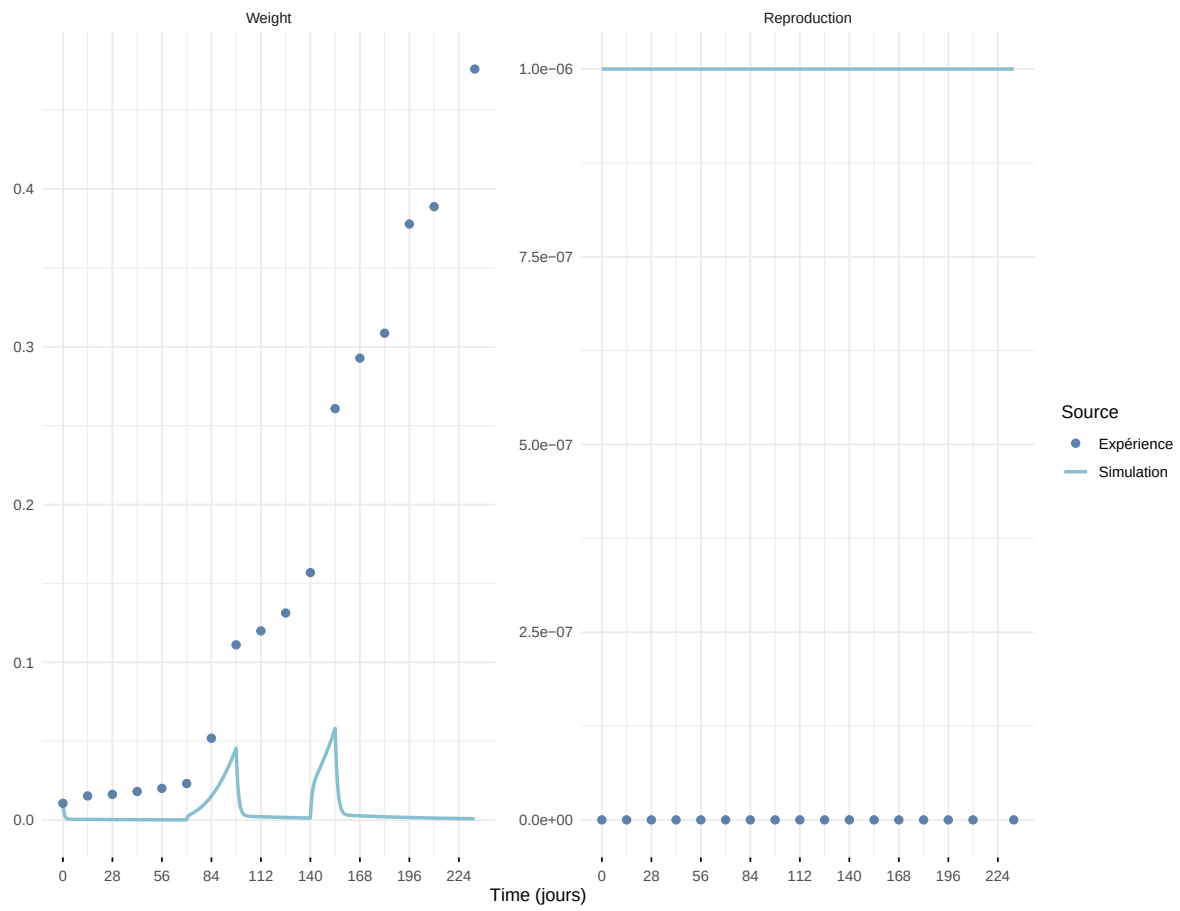
Expérience : Bart2019_XP1_1



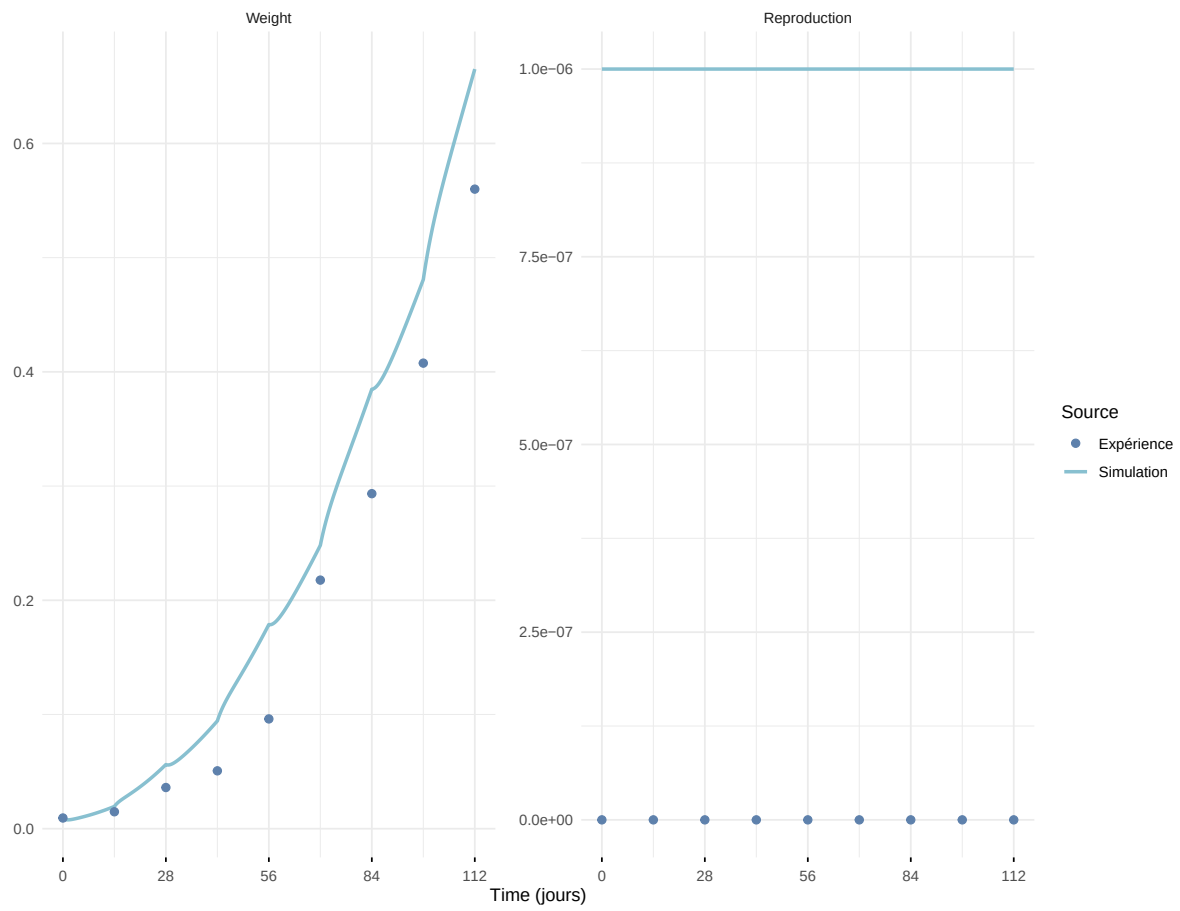
Expérience : Bart2019_XP1_2



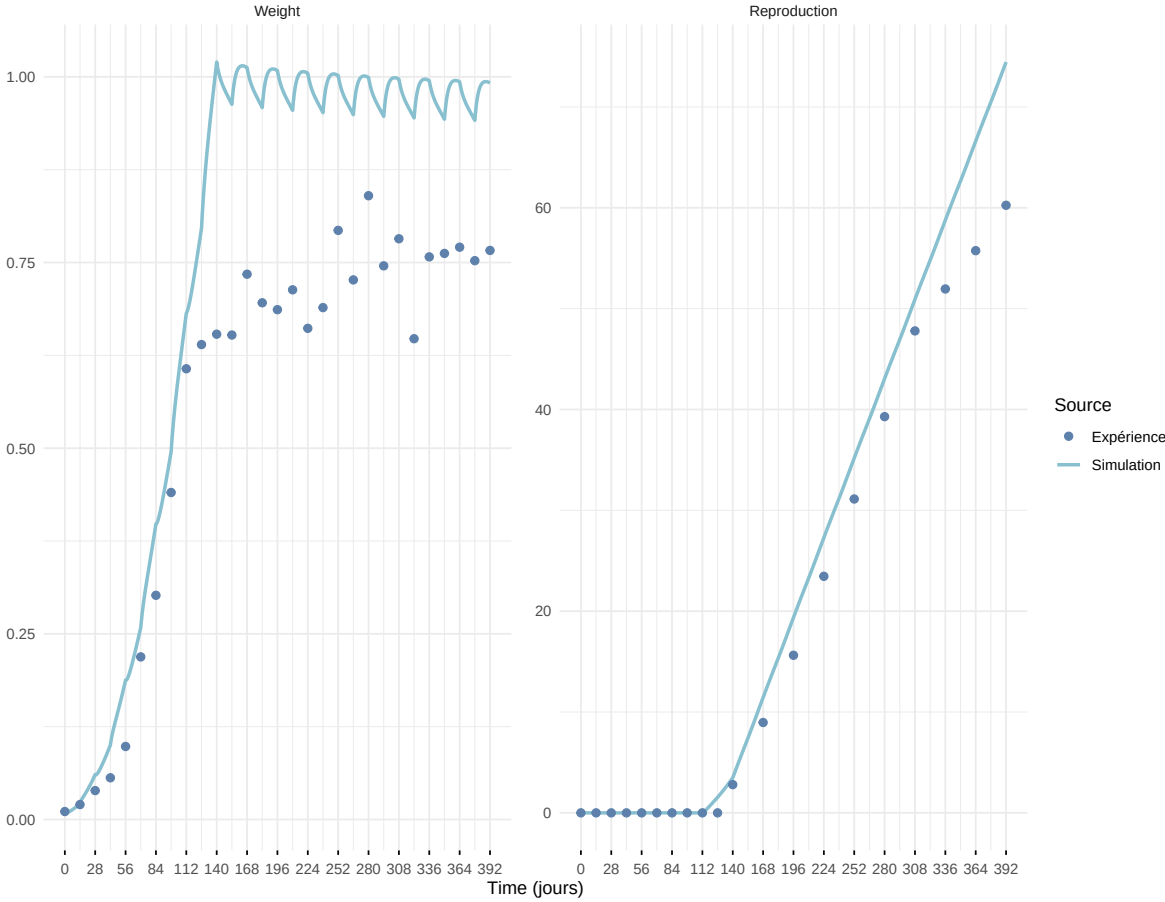
Expérience : Bart2019_XP3



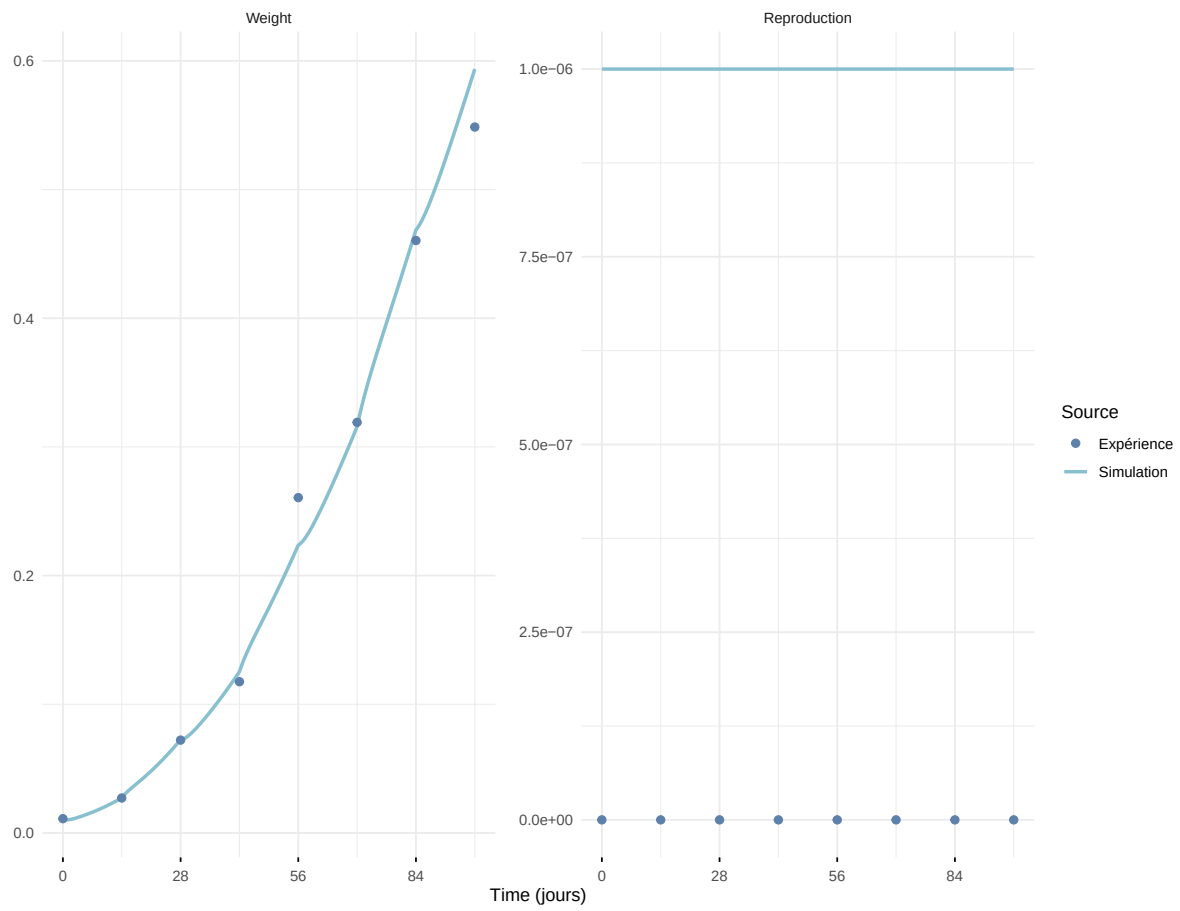
Expérience : Bart2019_XP4_1



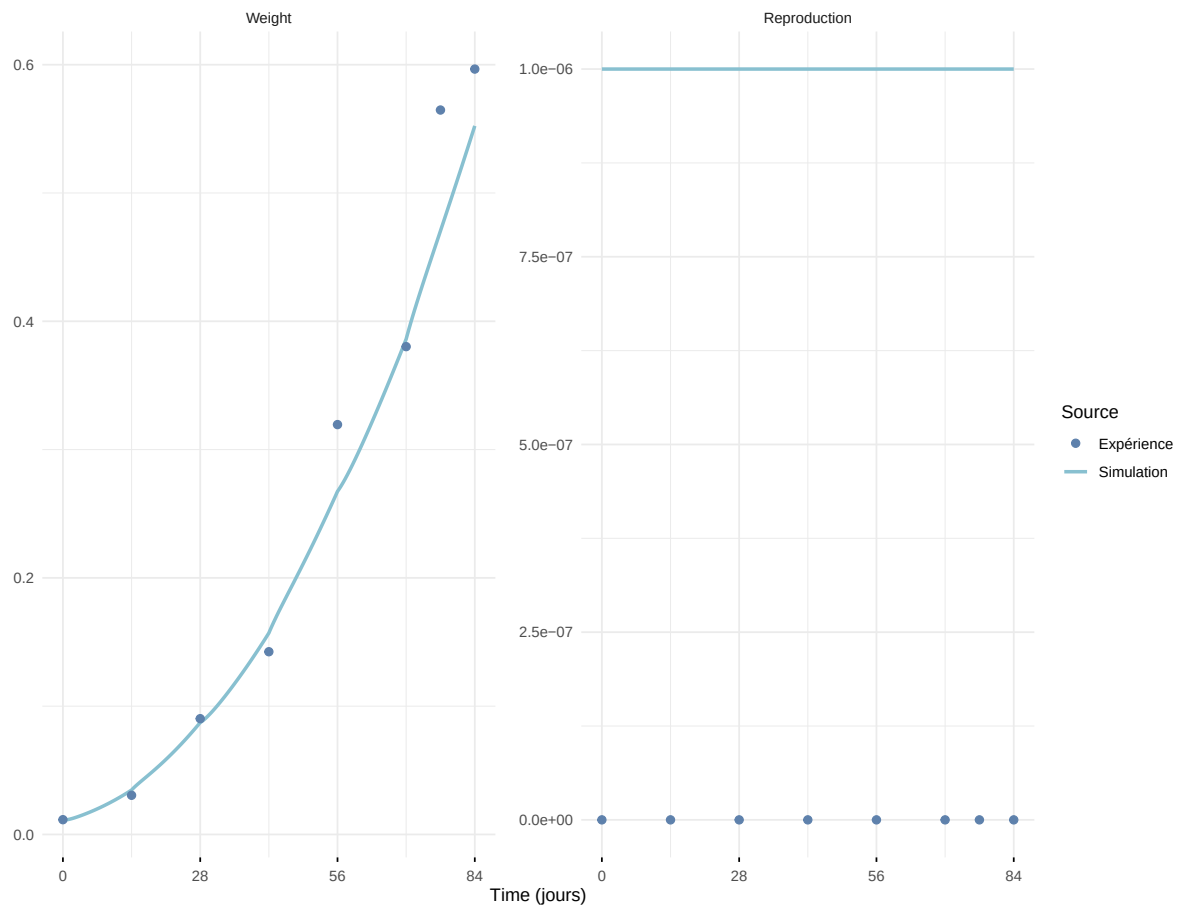
Expérience : Bart2019_XP4_2



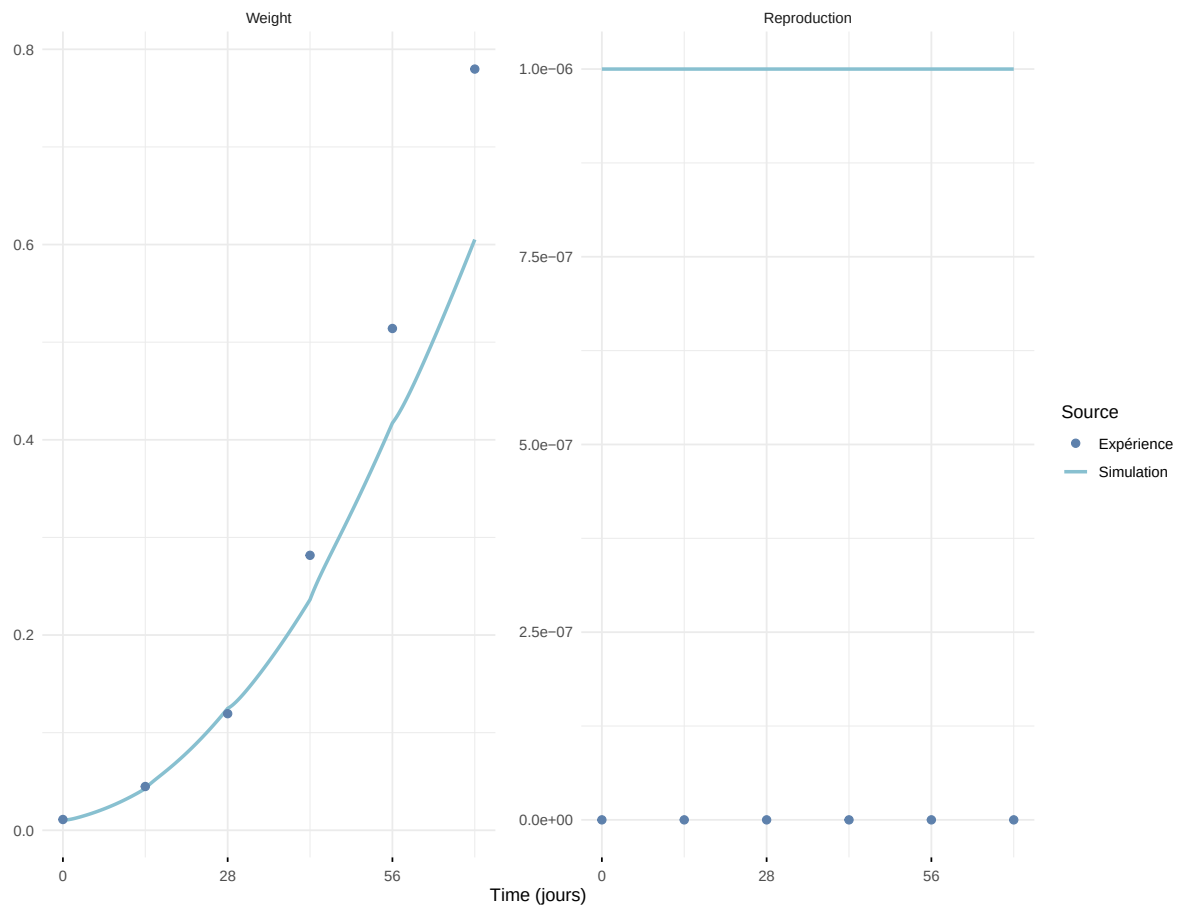
Expérience : Bart2019_XP5



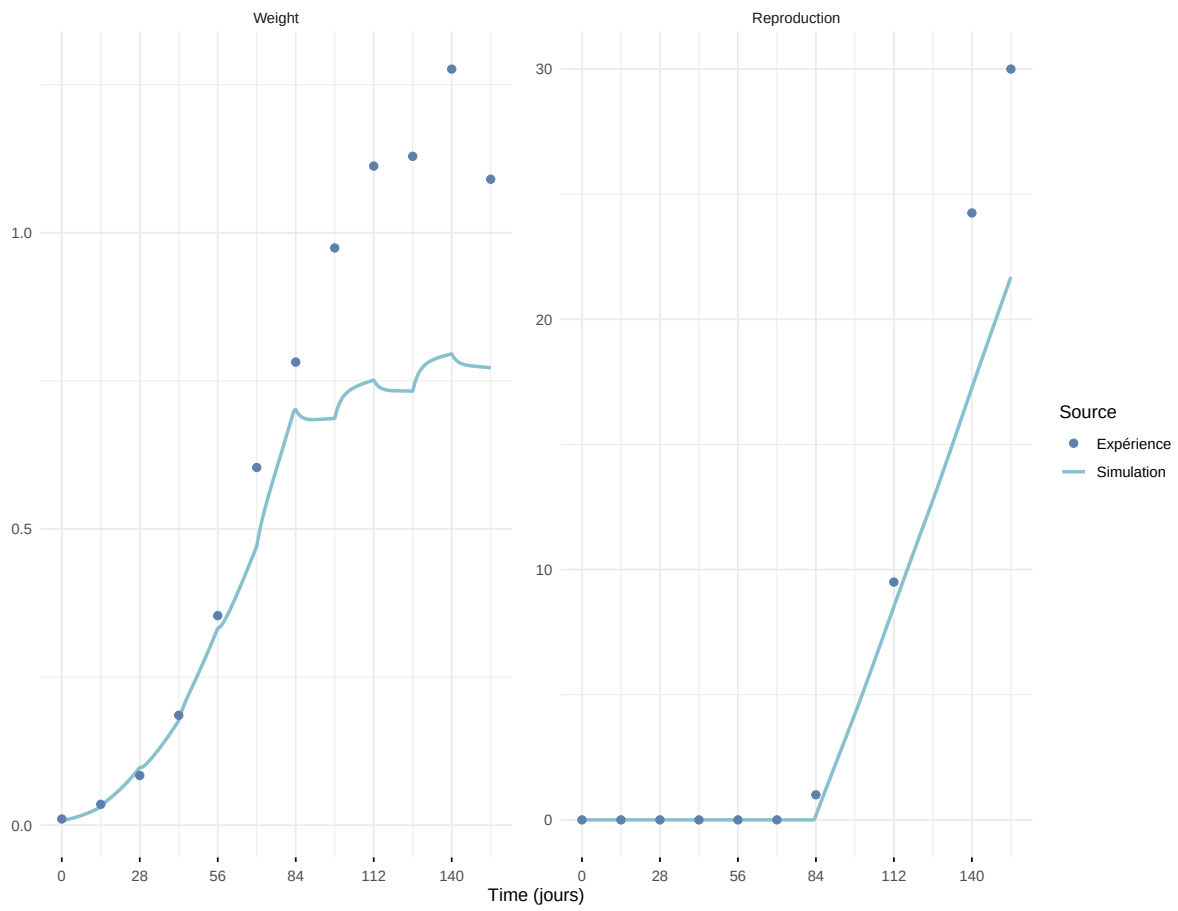
Expérience : Bart2020



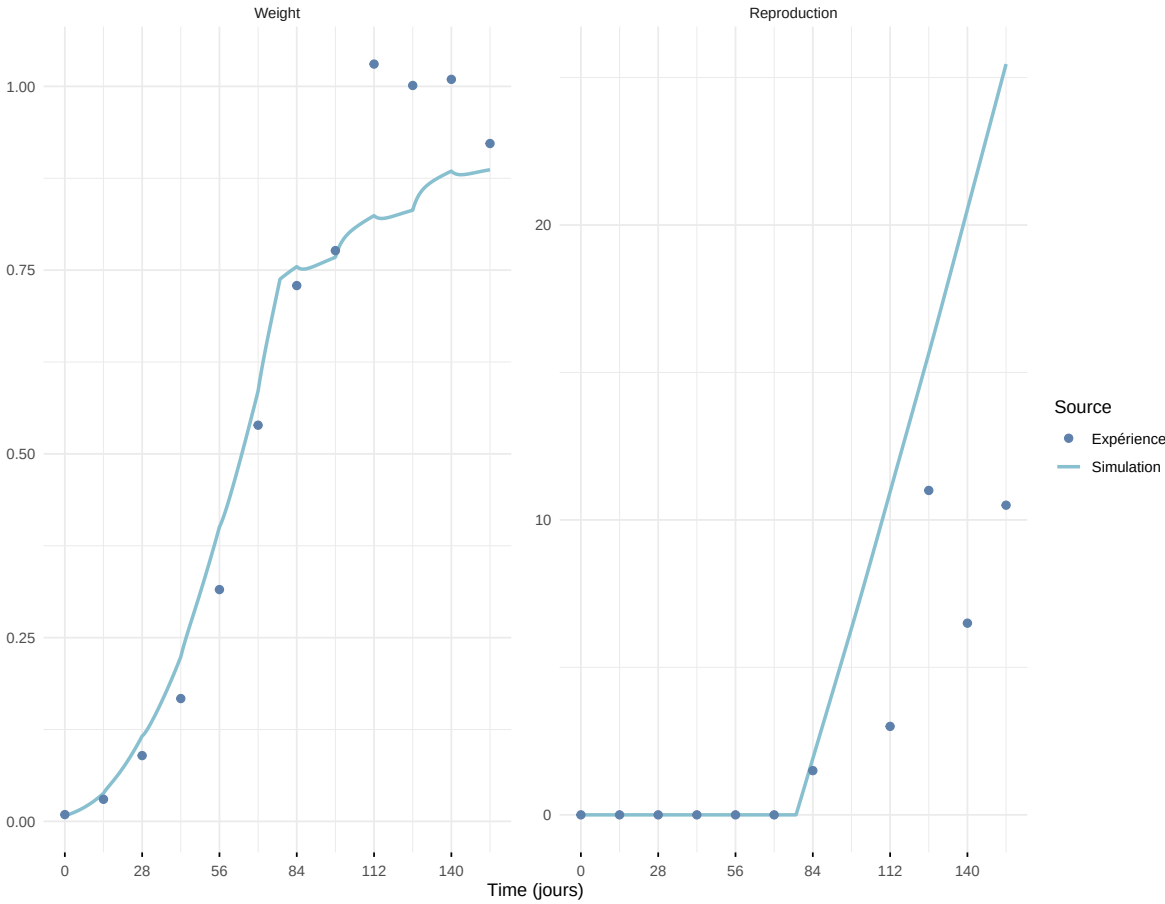
Expérience : Gollot2026_dens_D1



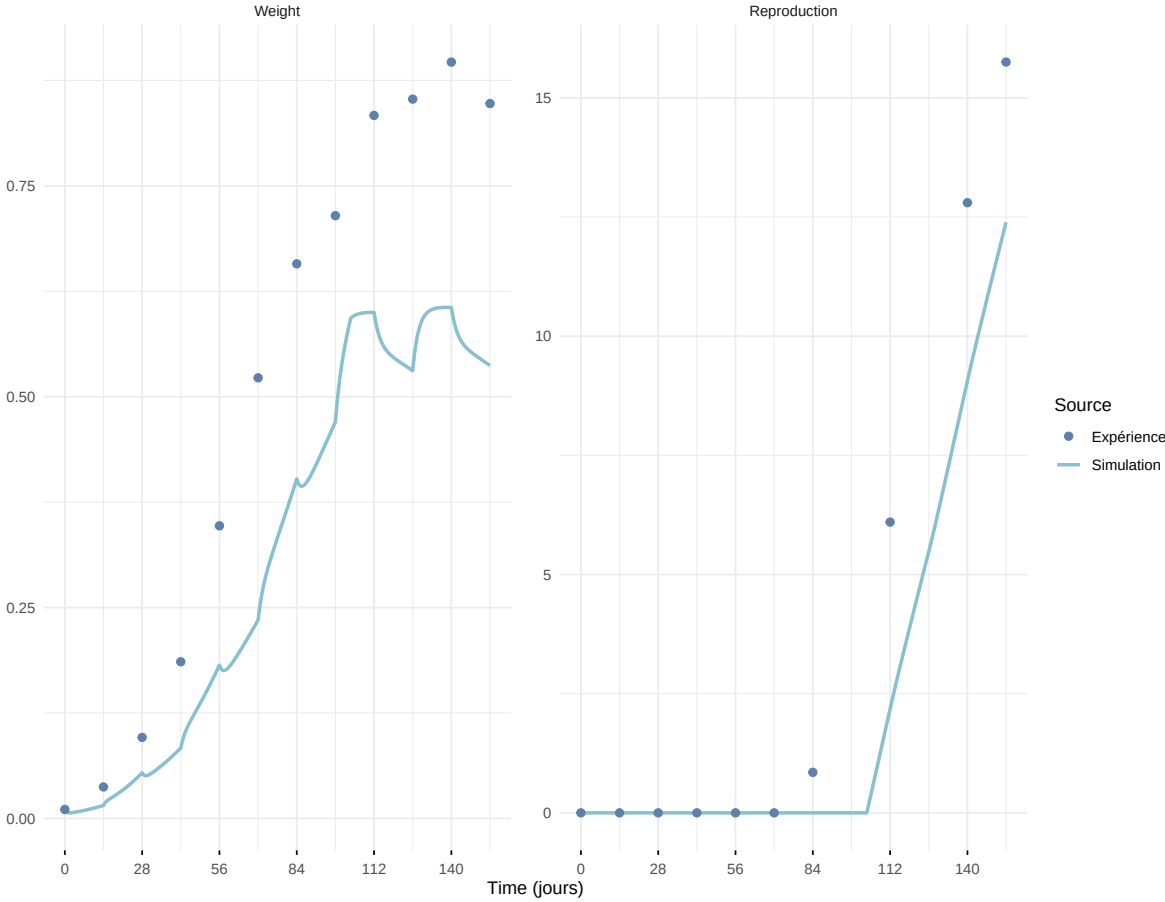
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

